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MEMO

TO: Portfolio Manager **FROM:** Quantitative Analysis Team **DATE:** January 12, 2026
SUBJECT: Risk Analysis and Valuation: AAPL Near-the-Money Call Options

Objective

This analysis evaluates the fair value and risk profile of the Apple Inc. (AAPL) call option maturing in approximately three months. Our goal is to determine if current market conditions justify a position and to define the necessary parameters for managing the associated directional and volatility risks.

Fair Value Assessment

Based on our Black-Scholes pricing engine using a historical volatility of 32.08%, the theoretical fair value of this call option is **\$17.67**.

It is critical to note that this is a model-driven estimate. While it provides a robust benchmark for "mathematical fairness" based on historical price fluctuations and current interest rates (4.5%), it does not account for upcoming binary events (e.g., earnings announcements) or sudden shifts in market sentiment that may cause market prices to deviate from historical norms.

Key Risks

Holding this position introduces three primary risk vectors that require constant oversight:

- **Directional Risk (Delta):** The position has a Delta of approximately 0.55. This means the option currently behaves like owning 55 shares of AAPL stock. You

benefit from price increases, but a

1 decline in AAPL shares will result in a loss of roughly \$55 per contract.

- **Time Decay (Theta):** Options are wasting assets. This specific contract loses approximately \$0.10 in value every single day due to the passage of time, regardless of stock price movement. This decay will accelerate as we approach the expiration date.
- **Volatility Exposure (Vega):** The position is sensitive to changes in market uncertainty. For every 1% drop in implied volatility, the option's value is expected to decrease by \$0.51 per contract. A "crush" in volatility following a major news event could erode gains even if the stock price moves in our favor.

Hedging & Monitoring Guidance

To manage the directional exposure, we recommend a Delta-hedging strategy. At current levels, selling **55 shares** of AAPL stock against each long call contract would neutralize the immediate impact of small stock price fluctuations, leaving the portfolio exposed primarily to changes in volatility and time.

We advise daily monitoring of the following:

1. **Spot Price vs. Strike:** To adjust the hedge ratio as Delta changes.
2. **Realized Volatility:** To ensure our 32% assumption remains valid.
3. **Time to Expiry:** To manage the increasing "cost of carry" from Theta decay.

Conclusion

The analysis suggests the option is fairly priced relative to its one-year historical volatility. We recommend a **Hedged Hold** stance. By neutralizing the Delta, the desk can capture potential gains from a rise in volatility or a significant upward breakout while remaining protected against minor downward drifts. Reassess the position if realized volatility exceeds 35% or if the stock price moves significantly away from the \$260 strike.